

HEMASHREE. P

Bachelor of Computer Science

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PROFESSIONAL SUMMARY

Aspiring AI researcher and B.Sc. Computer Science graduate with interests in Machine Learning, Deep Learning, Computer Vision, Visual AI Systems, and Multimodal AI. Experienced in conducting independent research and developing deep learning solutions for biomedical image analysis, explainable AI, and computer vision applications. Presented research on deep learning-based cell image classification at an international conference and completed an independent benchmark study evaluating the robustness of modern object detection models under motion blur. Passionate about building reliable AI systems, exploring emerging research directions, and contributing to meaningful advances in artificial intelligence through graduate-level research.

RESEARCH WORK

“Benchmarking Motion Blur Robustness in Modern Object Detectors: A Comparative Study of YOLOv8, RT-DETR, and GroundingDINO”

Independent Research | 2026

- Conducted a comparative robustness study of YOLOv8, RT-DETR, and GroundingDINO under motion blur degradation.
- Developed an evaluation pipeline using Python, PyTorch, OpenCV, and modern object detection frameworks.
- Proposed and analysed Detection Retention Rate (DRR) and Confidence Retention Rate (CRR) metrics.
- Evaluated models on COCO validation images with synthetically generated motion blur.
- Authored a research manuscript and released the project repository; arXiv preprint submission planned.

“Deep Learning-Based Classification of Cell Images for Biomechanical State Analysis”

Published in *Proceedings of the International Conference on Sustainable Material (ICSM 2025)*
Tamil Nadu State Council for Science and Technology - July 2025

- Developed a Convolutional Neural Network (CNN) using transfer learning (ResNet50 architecture) for biomechanical state classification of cell images.

- Implemented Grad-CAM for model interpretability and feature visualization.
- Optimized pre-processing and model fine-tuning to improve classification performance.
- Contributed to experimental analysis and technical documentation for publication.

TECHNICAL SKILLS

- **AI/ML & Computer Vision:** Machine Learning, Deep Learning, Convolutional Neural Networks (CNNs), Transfer Learning (ResNet50), Computer Vision, Biomedical Image Classification, Explainable AI (Grad-CAM), Image Processing, Data Preprocessing, Feature Extraction, Model Evaluation
- **Programming Languages:** Python, Java, C, R
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering
- **Backend & Web Development:** Flask, API Integration, HTML, CSS, Basic JavaScript
- **Databases:** SQL, MySQL
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle
- **Currently Learning:** TensorFlow, Keras, PyTorch, Advanced Deep Learning, Representation Learning, Generative AI
- **Other Skills:** Debugging, Analytical Thinking, Problem Solving, Technical Documentation, Research Presentation, Team Collaboration

PROJECTS.

ResearchMind: A Retrieval-Augmented Research Paper Assistant for Literature Review and Trend Analysis

- Built a Python-based research assistant using arXiv API, Sentence Transformers, and FAISS for academic paper retrieval and semantic search.
- Implemented a RAG pipeline for question answering, literature summarization, and research trend analysis.
- Developed automated report generation from retrieved research papers.
- Applied NLP, information retrieval, vector embeddings, and LLM concepts for research exploration.

Leukemia Cell Classification Using ResNet50 and Grad-CAM

- Developed a deep learning-based leukemia cell image classification model using ResNet50 transfer learning
- Applied Grad-CAM visualization for explainable AI and model interpretability
- Performed image preprocessing, augmentation, model training, and evaluation
- Worked with biomedical image datasets using computer vision techniques
- Technologies: Python, TensorFlow/Keras, OpenCV, NumPy, Matplotlib, Scikit-learn

Chest X-Ray Medical Image Analysis

- Created and documented an end-to-end machine learning workflow on Kaggle
- Included dataset preprocessing, visualization, model training, and evaluation
- Added markdown explanations and result analysis for better interpretability
- Demonstrated practical implementation of deep learning and computer vision concepts
- Used Python and data visualization libraries.

OTSeg-inspired Multi-Prompt CLIP Prototype

- Developed a CLIP-based prototype inspired by the OTSeg paper for zero-shot semantic segmentation.
- Implemented multi-prompt evaluation to compare and rank text prompts against image embeddings using similarity scores.
- Explored vision-language alignment and foundational concepts of multimodal attention for segmentation-style tasks.

Food Waste Reduction Web Application

- A Flask-based web application connecting restaurants and NGOs to minimize food waste.
- Implemented secure user registration, donation tracking, and real-time update features.
- Designed backend database integration for efficient management of donation records.

ACHIEVEMENTS & AWARDS

- **Debugging Competition** - Recognized for optimizing complex code under time constraints January 2024
- **Debugging Competition** - Awarded for analytical problem-solving and programming precision January 2025
- **Paper Presentation** on Artificial Neural Networks - February 2026

CERTIFICATIONS

- Smart Agriculture Monitoring System for Tamil Nadu Farmers - **Tamil Nadu Innovation Initiatives Hackathon, March 2026**
- **Python - Object-Oriented Programming (Infosys)** - Hands-on training on Python fundamentals and OOP concepts.
- **Frontend Development & Cloud Certification (IBM)** - Completed IBM-certified training in web development and cloud deployment.

INTERNSHIP

Web Development Intern - Velgrow Academy

- Developed a Python-based Nutrition Tracking Application using CustomTkinter.
- Implemented user authentication, nutrition logging, and diet recommendation features.
- Integrated backend logic with database management for efficient data storage and retrieval.
- Enhanced user experience through structured GUI design and responsive interface development.

EDUCATION

Bachelor of Science in Computer Science (2023-2026)

Queen Mary's College, Affiliated with the University of Madras, Chennai

VOLUNTEER EXPERIENCE

- **Community Service - Tree Planting Initiative:** Participated in local sustainability drives promoting environmental awareness.
- **Computer Knowledge Instructor** - Taught basic computer skills to school students.
- **College Volunteer (2023–2026)** - Actively contributed to organizing academic and extracurricular events. Collaborated with peers in coordinating college-level activities